Juan de Dios Alvarez

Data Scientist -- Predictive Modeling, Analytics and AI

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Profile

Data scientist with seven years of industry work (2019 to present), four of them delivering analytics and machine learning to US-based clients. I turn scattered data -- public statistics, operational records, documents -- into models, dashboards and reports that business teams use to decide. My work covers the full path: extraction and modeling in SQL, analysis in Python (pandas, scikit-learn) and R, testing each result against a baseline or ground truth before it is relied on, and presentation for non-technical readers in Power BI, Tableau or Superset. I also build AI and NLP components (LLM pipelines, retrieval, text classification) where they add measurable value.

Selected Results

- **Data analysis:** at FIRA, a development bank, automated the data behind credit-assignment and non-payment-risk analysis: 74 scheduled extraction scripts over 9+ public sources and 60+ SQL functions computing economic indicators.
- **Predictive models:** built forecasting models to identify consumer segments for client sales and marketing strategy (D3Clarity); built a crop-production estimation pipeline with Random Forest and Gradient Boosting compared by 5-fold cross-validation (FIRA).
- **Dashboards for the business:** Power BI dashboard that told a US client's sales team which products to offer each school district; 15+ Apache Superset dashboards and automated executive reports at FIRA.
- **Validating before relying on a result:** compared extraction methods by character and word error rates against ground truth; measured retrieval by recall@k against a labeled question bank, with a stratified sample to estimate misses.

Professional Experience

Data Scientist & ML Engineer | Independent Practice

March 2026 -- present

- Designed and run in production the data system of a small retail business in Mexico: PostgreSQL warehouse (40+ tables, PL/pgSQL) integrating point-of-sale, supplier, e-invoice and bank data; FIFO inventory valuation and bank-to-invoice reconciliation.
- Evaluated document-extraction methods (classical OCR, layout detectors, local vision and language models) head to head; adopted changes only on measured before/after results (doubtful lines reduced from 175 to 77 on a fixed sample).
- Built a regulatory-review method for anti-money-laundering compliance over 26 instruments: labeled control cases with an explicit acceptance criterion and stratified sampling to estimate misses.
- Built LLM pipelines with mechanical guardrails: schema-constrained outputs, rejection of text not present in the source, and local models for sensitive documents.

Economic Research Specialist (Data Scientist) | FIRA

January 2025 -- January 2026

- Replaced one analyst's manual Excel process with a scheduled, unattended pipeline: 74 extraction scripts (Python, Playwright, aiohttp) across 9+ public sources (INEGI, Banxico, SNIIM, SIAP, USDA) loading PostgreSQL with provenance metadata; 60+ PostgreSQL functions and materialized views computing inflation contributions, trade balances and supply-and-demand balances.
- Built a crop-production estimation pipeline from Sentinel-2 vegetation-index time series for 310 plots (pandas, scikit-learn): outlier filtering, PCA variable selection, and Random Forest and Gradient Boosting regressors compared by 5-fold cross-validation, trained on a proxy target because field-level yields were not available.

- Delivered automated executive Excel reports and 15+ Apache Superset dashboards to decision-makers; set governance conventions (raw data preserved, lineage, GitLab) and supervised an undergraduate team.

Data Scientist (contract) | CITEIM

October 2024 -- January 2025

- Built the dataset for a Mexican Sign Language recognition system: video sourcing, weak labels from subtitles (OCR and speech-to-text), MediaPipe keypoints.

Data Scientist & Team Lead | Punto Singular

December 2023 -- August 2024

- Led monthly analytics cycles for an EdTech client, from problem and metric definition to results presented against the agreed metric; KPI and quality dashboards in R, Python, Tableau and Looker used directly by business stakeholders.
- Founded and led a Data Science group of undergraduate students working on real client and nonprofit projects.

Data Scientist (contract) | BairesDev

September 2023 -- November 2023

- Built a retrieval-augmented generation proof of concept that let a legal client's lawyers query their contracts for obligations, risk clauses and precedents; also delivered transformer-based summarization and contract classification.

Data Scientist | D3Clarity (formerly Viqtor Davis North America)

February 2019 -- March 2023

- First data scientist of the company; defined the analytics roadmap and delivered it to US-based clients in a distributed team.
- Built forecasting models to identify consumer segments for client sales and marketing strategy.
- Classified public school-board meeting minutes with NLP on Azure ML, from proof of concept to production, and delivered a Power BI dashboard used by the client's sales team.
- Built executive and operational dashboards in Power BI and Tableau for non-technical decision-makers; data lakes and ETL.

Earlier Experience

- Lecturer, Tecnologico de Monterrey Business School (KPI dashboarding in Tableau for economics and management students); developer, University of Michoacan IT; HAWC observatory data analysis; data-acquisition software, CERN Crystal Clear Collaboration.

Education

- **Master's in Computer Science** -- INAOE, Mexico. Degree awarded; SEP professional license (cedula) 5916605. Thesis: automatic text classification using prototype-based class reduction.
- **PhD Candidate in Physics** -- University of Michoacan. Computational research: large-scale simulations of cosmic-ray air showers to search for signals of physics beyond the Standard Model; principal author of two ICRC 2017 papers; co-author on 14 publications.
- **Bachelor's in Physics and Mathematics** -- University of Michoacan.
- **Certifications:** AWS Certified Cloud Practitioner; Semarchy xDM. **Languages:** Spanish (native), English (professional), Italian (basic-intermediate), French (basic).

Technical Skills

- **Languages and analysis:** Python (pandas, NumPy, scikit-learn), SQL, R
- **Predictive modeling:** Random Forest, Gradient Boosting, PCA, forecasting, clustering, text classification
- **Hypothesis validation:** baselines, ground truth, k-fold cross-validation, recall@k, control sets, stratified sampling
- **Dashboards and reporting:** Power BI, Tableau, Apache Superset, Looker, automated Excel reports
- **Data and cloud:** PostgreSQL (PL/pgSQL, window functions, materialized views), MySQL; ETL, APIs; AWS, Azure ML
- **AI and NLP:** LLM pipelines (Claude, local models via Ollama), RAG, embeddings, schema-constrained extraction, NER